The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Section-Centered Hidden-Response
Observer-Relevant Pullback Body-Response Core
from a Section-Centered Displacement-Response Substrate Law

Abstract

The current active-primary theorem already proved that the observer-relevant pullback body-response core $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,bodyresp}}$ is the unique minimal benchmark-facing datum on the recorded section-centered hidden-response route up to body-response-faithful equivalence. The next honest benchmark-facing move is therefore no longer another same-core relay. It is to derive that current core itself, or some nontrivial constituent inside it, from still more primitive explicit substrate structure.

The present note makes that move directly. For each sector it introduces a benchmark-faithful section-centered displacement-response substrate law consisting of a compact convex section-centered substrate body over the fixed proto-section body, a surjective displacement projection to the recorded section-centered displacement space, a positive self-adjoint substrate response operator, and a K-orthogonal minimal-lift splitting along the displacement projection. The current section-centered displacement-body datum is then realized as the projection image of the substrate body, while the body-restricted pullback response is no longer primitive: it is the unique minimal substrate-response energy among all substrate lifts of a given recorded displacement. In this sense the current body-response core is forced by a more primitive explicit substrate law rather than merely postulated.

The benchmark boundary remains fixed throughout. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, the low-energy matter law is not enlarged, and no stronger promotion language is licensed. By the earlier collapse theorem and the current necessity / uniqueness frontier, the same pullback body-operator core, pullback-operator core, pullback-quadratic core, section-centered residual-normal-form realization, continuity-Hessian compressed core, and benchmark one-metric observer package then follow unchanged. The positive result is therefore a genuine derivational gain below the body-response-core necessity / uniqueness frontier rather than another same-output deeper relay.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity now has a sharper burden than before. The current body-response-core necessity / uniqueness theorem already showed that, on the recorded section-centered hidden-response route, the observer-relevant pullback body-response core $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,bodyresp}}$ is the unique minimal benchmark-facing datum up to body-response-faithful equivalence. That theorem therefore spent the option of merely repackaging the same

body-response core or re-expanding back to the larger pullback-body-operator, pullback-operator, pullback-quadratic, hidden-response, residual-normal-form, or pre-proto-section presentations.

What remains honest is to derive that current core itself from more primitive explicit substrate structure, or else to prove a genuinely smaller theorem inside it. The present note takes the direct derivation route. It does not reintroduce the larger hidden-response package as the live benchmark-facing datum. Instead it assigns the current displacement body as a projection image of a more primitive substrate body and derives the current body-restricted pullback response as the unique minimal quadratic substrate-response cost compatible with each recorded displacement. In that precise sense, the current body-response core is no longer primitive once the section-centered displacement-response substrate law is given.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It does not introduce a second operative metric, it does not enlarge the low-energy matter law, and it does not claim a completed first-principles derivation of GR from final substrate microphysics. Its narrower benchmark-facing role is to show that the current observer-relevant pullback body-response core is already forced by a smaller explicit substrate law tailored directly to the current section-centered body-response frontier.

2. Fixed Inputs from the Current Body-Response Frontier

Definition 1 (Recorded primitive continuation data)**.** *Fix the following continuation data from the active primitive-reservoir observer-reduction line.*

1. *For each sector label* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a primitive forcing shell*

$$\mathcal{I}_\sigma^{\mathrm{for}}.$$

2. *The product shell*

$$\mathcal{S}_{\mathrm{prim}} \equiv \mathcal{I}_{\mathrm{car}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{cur}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{hom}}^{\mathrm{for}}.$$

3. *The recorded metric and ordinary-matter outputs*

$$\mathscr{G}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \to \{\textit{observer metrics}\}, \qquad \mathscr{T}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \times \Psi \to \{\textit{observer stress tensors}\},$$

satisfying

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^\sharp, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T_{\mu\nu}^{\mathrm{matter}}[g^\sharp, \psi] \tag{1}$$

for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ *and every matter configuration* $\psi \in \Psi$*.*

Definition 2 (Recorded benchmark base and downstream chain)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. The active line already fixes the following data and consequences.*

1. *the current proto-section benchmark base* $\mathfrak{Q}^{\mathrm{sub,proto,secgen}}$*, in particular the recorded proto-section body* $\mathcal{B}_\sigma^{\mathrm{psec}}$ *and shell* $\mathcal{Z}_\sigma^{\mathrm{psec}}$*;*

2. *the current observer-relevant pullback body-response core* $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,bodyresp}}$*, the current observer-relevant pullback body-operator core* $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,bodyop}}$*, the current observer-relevant pullback-operator core* $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,op}}$*, the current observer-relevant pullback-quadratic core* $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb}}$*, the current section-centered residual-normal-form realization* $\mathfrak{R}^{\mathrm{sub,proto,presec,res}}$*, the current observer-relevant pre-proto-section minimizer-section core* $\mathfrak{R}^{\mathrm{sub,proto,presec,sec}}$*, and the current benchmark-faithful pre-proto-section package* $\mathfrak{R}^{\mathrm{sub,proto,presec}}$*;*

3. *the previously recorded fact that, once a benchmark-faithful observer-relevant pullback body-response core over the fixed proto-section benchmark base is available, the current body-operator-core realization, the current pullback-operator-core realization, the current pullback-quadratic-core realization, the current section-centered residual-normal-form realization, the current pre-proto-section package, the current minimizer-section core, and the previously recorded continuity-Hessian compressed core and benchmark one-metric observer-package consequences follow unchanged;*

4. *benchmark faithfulness, meaning preservation of the benchmark outputs* (1)*, no second operative metric, no enlarged low-energy matter law, no change to the ordinary matter route, and no premature promotion from adoption to completed derivation.*

Definition 3 (Current observer-relevant pullback body-response core)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. A* benchmark-faithful observer-relevant pullback body-response core $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,bodyresp}}$ *for sector* σ *consists of:*

1. *a compact convex section-centered displacement body*

$$\mathcal{D}_\sigma^{\mathrm{pppresec}} \subset \mathcal{B}_\sigma^{\mathrm{psec}} \times V_\sigma^{\mathrm{pppresec}}$$

   whose fiber

$$\mathcal{D}_\sigma^{\mathrm{pppresec}}(y) \equiv \{v \in V_\sigma^{\mathrm{pppresec}} : (y, v) \in \mathcal{D}_\sigma^{\mathrm{pppresec}}\}$$

   is nonempty, compact, convex, contains 0*, and has nonempty interior for every* $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$*;*

2. *the body-restricted pullback response*

$$\mathscr{R}_\sigma^{\mathrm{pppresec,pb,body}} : \mathcal{D}_\sigma^{\mathrm{pppresec}} \to [0, \infty),$$

   smooth in y*, fiberwise quadratic in* v*, and satisfying*

$$\mathscr{R}_\sigma^{\mathrm{pppresec,pb,body}}(y, v) = 0 \qquad \Longleftrightarrow \qquad v = 0$$

   on every fiber;

3. *benchmark faithfulness in the sense of Definition 2(4).*

Proposition 1 (What the current frontier already fixes)**.** *On the recorded section-centered hidden-response route, the current frontier already fixes two benchmark-facing facts.*

1. *The observer-relevant pullback body-response core* $\mathfrak{S}^{\mathrm{sub,proto,ppresec,pb,bodyresp}}$ *is the unique minimal benchmark-facing datum at current scope up to body-response-faithful equivalence.*

2. *All current benchmark-facing downstream uses already factor through the current body-response core and the fixed proto-section benchmark base.*

Proof. Item (1) is exactly the current body-response-core necessity / uniqueness theorem. Item (2) is exactly the previously recorded collapse of the observer-relevant pullback body-operator core to the smaller observer-relevant pullback body-response core. □

3. Section-Centered Displacement-Response Substrate Law

Definition 4 (Benchmark-faithful section-centered displacement-response substrate law)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. A* benchmark-faithful section-centered displacement-response substrate law *for sector* σ *consists of the following data.*

1. *A finite-dimensional Euclidean substrate carrier*

$$U_\sigma^{\mathrm{disp,resp}}$$

and a compact convex section-centered substrate body

$$\mathcal{U}_\sigma^{\mathrm{disp,resp}} \subset \mathcal{B}_\sigma^{\mathrm{psec}} \times U_\sigma^{\mathrm{disp,resp}}$$

whose fiber

$$\mathcal{U}_\sigma^{\mathrm{disp,resp}}(y) \equiv \left\{ u \in U_\sigma^{\mathrm{disp,resp}} : (y,u) \in \mathcal{U}_\sigma^{\mathrm{disp,resp}} \right\}$$

is nonempty, compact, convex, contains 0*, and has nonempty interior for every* $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$*.*

2. *A smooth family of surjective linear displacement projections*

$$P_\sigma^{\mathrm{disp,resp}}(y) : U_\sigma^{\mathrm{disp,resp}} \to V_\sigma^{\mathrm{ppresec}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}}.$$

3. *A smooth family of positive self-adjoint substrate response operators*

$$K_\sigma^{\mathrm{disp,resp}}(y) : U_\sigma^{\mathrm{disp,resp}} \to U_\sigma^{\mathrm{disp,resp}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}},$$

satisfying the uniform coercive bound

$$\left\langle u, K_\sigma^{\mathrm{disp,resp}}(y) u \right\rangle \geq \mu_\sigma \|u\|^2 \qquad \text{for all } y \in \mathcal{B}_\sigma^{\mathrm{psec}}, \quad u \in U_\sigma^{\mathrm{disp,resp}} \tag{2}$$

for some constant $\mu_\sigma > 0$*.*

4. *A smooth family of linear minimal lifts*

$$J_\sigma^{\mathrm{disp,resp}}(y) : V_\sigma^{\mathrm{ppresec}} \to U_\sigma^{\mathrm{disp,resp}}$$

satisfying, for every $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$*,*

(a)

$$P_\sigma^{\mathrm{disp,resp}}(y) \circ J_\sigma^{\mathrm{disp,resp}}(y) = \mathrm{id}_{V_\sigma^{\mathrm{ppprese c}}};$$

(b) every $u \in U_\sigma^{\mathrm{disp,resp}}$ *admits the unique decomposition*

$$u = J_\sigma^{\mathrm{disp,resp}}(y)\big(P_\sigma^{\mathrm{disp,resp}}(y)u\big) + k, \qquad k \in N_\sigma^{\mathrm{disp,resp}}(y) \equiv \ker P_\sigma^{\mathrm{disp,resp}}(y);$$

(c) the lift is $K_\sigma^{\mathrm{disp,resp}}(y)$*-orthogonal to the kernel, namely*

$$\left\langle J_\sigma^{\mathrm{disp,resp}}(y)v, K_\sigma^{\mathrm{disp,resp}}(y)k \right\rangle = 0$$

for every $v \in V_\sigma^{\mathrm{ppprese c}}$ *and every* $k \in N_\sigma^{\mathrm{disp,resp}}(y)$*.*

5. *Exact-shell discipline, meaning that the projected zero slice of the substrate body records exactly the zero section over the fixed proto-section shell:*

$$\left\{ y \in \mathcal{B}_\sigma^{\mathrm{psec}} : \exists u \in \mathcal{U}_\sigma^{\mathrm{disp,resp}}(y) \text{ with } P_\sigma^{\mathrm{disp,resp}}(y)u = 0 \right\} = \mathcal{Z}_\sigma^{\mathrm{psec}}.$$

6. *Benchmark faithfulness in the sense of Definition 2(4).*

Definition 5 (Current-body-realizing law)**.** *A benchmark-faithful section-centered displacement-response substrate law for sector* σ *is* current-body-realizing *if, for every* $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$*,*

$$\mathcal{D}_\sigma^{\mathrm{ppprese c}}(y) = P_\sigma^{\mathrm{disp,resp}}(y)\big(\mathcal{U}_\sigma^{\mathrm{disp,resp}}(y)\big). \tag{3}$$

Thus the current section-centered displacement-body datum is the projection image of the explicit substrate body.

Remark 1. *Definition 4 is deliberately smaller than the older full section-centered hidden-response package. It does not reintroduce a separate hidden-response space, a larger pullback-operator presentation, or any re-expansion back to the already spent hidden-response frontier. It is tailored only to the current body-response burden: force the recorded displacement body and the quadratic pullback response realized on that body.*

4. Forced Body-Restricted Pullback Response

Definition 6 (Substrate response energy and law-forced body response)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$ *and a benchmark-faithful section-centered displacement-response substrate law. Define the quadratic substrate response energy by*

$$\mathscr{E}_\sigma^{\mathrm{disp,resp}}(y,u) \equiv \frac{1}{2}\left\langle u, K_\sigma^{\mathrm{disp,resp}}(y)u \right\rangle.$$

For $v \in P_\sigma^{\mathrm{disp,resp}}(y)(U_\sigma^{\mathrm{disp,resp}})$*, define the* law-forced body response *by*

$$\widehat{\mathscr{R}}_\sigma^{\mathrm{disp,resp}}(y,v) \equiv \min\left\{ \mathscr{E}_\sigma^{\mathrm{disp,resp}}(y,u) : u \in U_\sigma^{\mathrm{disp,resp}},\ P_\sigma^{\mathrm{disp,resp}}(y)u = v \right\}. \tag{4}$$

Proposition 2 (Unique minimal substrate lift)**.** *Fix* $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$ *and* $v \in V_\sigma^{\mathrm{ppprese c}}$*. For every benchmark-faithful section-centered displacement-response substrate law, the constrained minimum*

in (4) *exists, is unique, and is achieved by the lifted substrate state*

$$u_{\min}(y,v) = J_\sigma^{\mathrm{disp,resp}}(y)v.$$

Consequently,

$$\widehat{\mathscr{R}}_\sigma^{\mathrm{disp,resp}}(y,v) = \frac{1}{2}\left\langle J_\sigma^{\mathrm{disp,resp}}(y)v, K_\sigma^{\mathrm{disp,resp}}(y)J_\sigma^{\mathrm{disp,resp}}(y)v\right\rangle. \tag{5}$$

Proof. Let $u \in U_\sigma^{\mathrm{disp,resp}}$ satisfy $P_\sigma^{\mathrm{disp,resp}}(y)u = v$. By Definition 4(4), u admits the unique decomposition

$$u = J_\sigma^{\mathrm{disp,resp}}(y)v + k, \qquad k \in N_\sigma^{\mathrm{disp,resp}}(y).$$

Using the $K_\sigma^{\mathrm{disp,resp}}(y)$-orthogonality of the lift to the kernel,

$$\left\langle u, K_\sigma^{\mathrm{disp,resp}}(y)u\right\rangle = \left\langle J_\sigma^{\mathrm{disp,resp}}(y)v, K_\sigma^{\mathrm{disp,resp}}(y)J_\sigma^{\mathrm{disp,resp}}(y)v\right\rangle + \left\langle k, K_\sigma^{\mathrm{disp,resp}}(y)k\right\rangle.$$

By the coercive bound (2), the second term is nonnegative and vanishes only when $k = 0$. Hence the minimum among all lifts of v is attained uniquely at $k = 0$, namely at $u = J_\sigma^{\mathrm{disp,resp}}(y)v$. Formula (5) follows immediately. □

Proposition 3 (The law-forced response is a benchmark-faithful body response)**.** *Fix a benchmark-faithful current-body-realizing section-centered displacement-response substrate law for sector σ. Then the pair*

$$\left(\mathcal{D}_\sigma^{\mathrm{pppresec}}, \widehat{\mathscr{R}}_\sigma^{\mathrm{disp,resp}}\right)$$

is a benchmark-faithful observer-relevant pullback body-response core over the fixed proto-section benchmark base.

Proof. Because the law is current-body-realizing, the recorded displacement body is exactly the projection image (3) of the compact convex substrate body. Hence each fiber $\mathcal{D}_\sigma^{\mathrm{pppresec}}(y)$ is the image of a nonempty compact convex set under a linear map, and so remains nonempty, compact, and convex; its nonempty interior and zero section are already part of the recorded current-body datum. By Proposition 2, the forced response $\widehat{\mathscr{R}}_\sigma^{\mathrm{disp,resp}}(y,v)$ is a smooth fiberwise quadratic form in v. If $v \neq 0$, then $J_\sigma^{\mathrm{disp,resp}}(y)v \neq 0$ because $P_\sigma^{\mathrm{disp,resp}}(y) \circ J_\sigma^{\mathrm{disp,resp}}(y) = \mathrm{id}$, and the positivity of $K_\sigma^{\mathrm{disp,resp}}(y)$ gives

$$\widehat{\mathscr{R}}_\sigma^{\mathrm{disp,resp}}(y,v) > 0.$$

Conversely $\widehat{\mathscr{R}}_\sigma^{\mathrm{disp,resp}}(y,0) = 0$. Benchmark faithfulness is part of Definition 4(6). Therefore the induced pair satisfies Definition 3. □

5. Forcing the Current Body-Response Frontier

Theorem 1 (Current body-response core from a section-centered displacement-response substrate law)**.** *Fix the primitive continuation data of Definition 1, the recorded proto-section benchmark base and downstream chain of Definition 2, and for each sector $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$ a benchmark-faithful current-body-realizing section-centered displacement-response substrate law in the sense of Definitions 4 and 5. Then:*

1. *the current section-centered displacement-body datum is the projection image of the explicit substrate body by* (3)*;*

2. *the body-restricted pullback response on that current displacement body is forced by the law as the unique minimal substrate-response energy* $\widehat{\mathscr{R}}_{\sigma}^{\mathrm{disp,resp}}$ *of Definition 6;*

3. *the pair*
$$\left(\mathcal{D}_{\sigma}^{\mathrm{pppresec}}, \widehat{\mathscr{R}}_{\sigma}^{\mathrm{disp,resp}}\right)$$
is therefore a benchmark-faithful observer-relevant pullback body-response core by Proposition 3;

4. *by Proposition 1(1), this law-forced core is body-response-faithfully equivalent to the unique minimal benchmark-facing body-response datum on the recorded route;*

5. *by Proposition 1(2), the same current pullback body-operator core, current pullback-operator core, current pullback-quadratic core, current section-centered residual-normal-form realization, current pre-proto-section package, continuity-Hessian compressed core, and benchmark one-metric observer package follow unchanged.*

Proof. Item (1) is Definition 5. Item (2) is Proposition 2. Item (3) is Proposition 3. For item (4), Proposition 1(1) says that any benchmark-faithful observer-relevant pullback body-response core on the current route is body-response-faithfully equivalent to the unique minimal datum already fixed by the current frontier theorem. Item (5) is then exactly Proposition 1(2): once a benchmark-faithful body-response core over the fixed proto-section benchmark base is in hand, all current downstream benchmark-facing consequences factor through it. □

Corollary 1 (The current body-response core is no longer primitive)**.** *Under the hypotheses of Theorem 1, the current observer-relevant pullback body-response core is no longer a primitive stopping point on the recorded section-centered hidden-response route. It is a forced output of the benchmark-faithful section-centered displacement-response substrate law.*

Proof. Theorem 1(1)–(3) shows that the displacement-body datum and body-restricted pullback response are supplied by the law rather than postulated independently. The current core is therefore induced by more primitive explicit substrate structure. □

Corollary 2 (Primary post-substrate-law burden)**.** *After Theorem 1, the next primary manuscript below the current frontier must do at least one of the following:*

1. *derive or force the section-centered displacement-response substrate law itself from still more primitive explicit substrate structure;*

2. *derive or force a nontrivial constituent of that law, such as the substrate body, the displacement projection / minimal-lift splitting, or the coercive substrate-response operator;*

3. *prove a sharper necessity, uniqueness, or compatibility theorem for a nontrivial constituent or relation inside that law;*

4. *do one of the previous three while preserving the same benchmark one-metric package, the same ordinary matter route, and the same claim boundary.*

Another same-output relay that merely preserves the same displacement-response substrate law, or any re-expansion back to the larger pullback-body-operator, pullback-operator, pullback-quadratic, hidden-response, residual-normal-form, or pre-proto-section presentations, is not the honest default next move.

Proof. By Corollary 1, the current body-response core is no longer the live unexplained stopping point. The remaining honest burden therefore moves upstream to the law that now forces that core, or to a strictly smaller theorem inside that law. □

Proposition 4 (The benchmark package remains fixed)**.** *The present substrate-law theorem preserves the benchmark one-metric package* (1)*. In particular, the operative metric remains exactly* $g^\sharp$*, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged.*

Proof. This is part of benchmark faithfulness in Definitions 2(4) and 4(6). The present note changes only the upstream status of the current body-response core on the recorded route. □

6. Conclusion

The positive result of this note is the direct derivation the routing layer asked for. On the recorded section-centered hidden-response route, the current section-centered displacement-body datum is now realized as the projection image of a more primitive explicit substrate body, and the current body-restricted pullback response is forced as the unique minimal substrate-response energy compatible with each recorded displacement.

That result is narrower than a completed first-principles derivation of GR from substrate microphysics, but it is stronger than another same-output deeper relay. The current observer-relevant pullback body-response core is no longer merely the unique minimal benchmark-facing datum on the route. It is now a forced output of the section-centered displacement-response substrate law. By the earlier collapse theorem and the current necessity / uniqueness frontier, the same pullback body-operator core, pullback-operator core, pullback-quadratic core, section-centered residual-normal-form realization, continuity-Hessian compressed core, and benchmark one-metric observer package therefore follow unchanged.

The scope remains honest. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, the low-energy matter law is not enlarged, and no stronger promotion language is licensed. The next open burden is therefore no longer the body-response core itself. It is to derive or force the section-centered displacement-response substrate law, or some nontrivial constituent or relation inside it, from still more primitive explicit substrate structure.